

AMD MICROBLAZE™ V PROCESSOR INTRODUCTION

*Embracing RISC-V with
AMD MicroBlaze V*

October 2024

AMD 
together we advance_

WHAT IS RISC-V?



An Instruction Set Architecture (ISA)

- **Not a processor** – it's a specification
- Controlled by RISC-V International

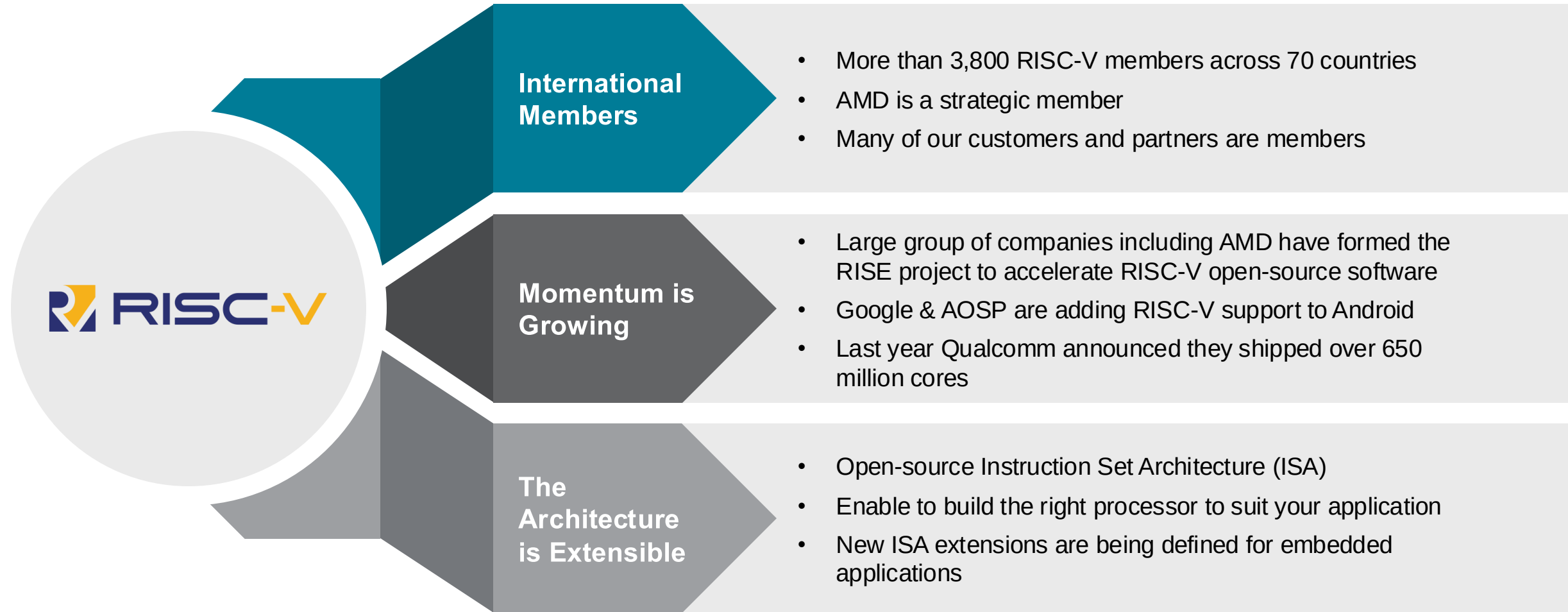
Open-Standard

- Anyone can develop a RISC-V processor – no licensing fees
- Implementation and innovation opportunities

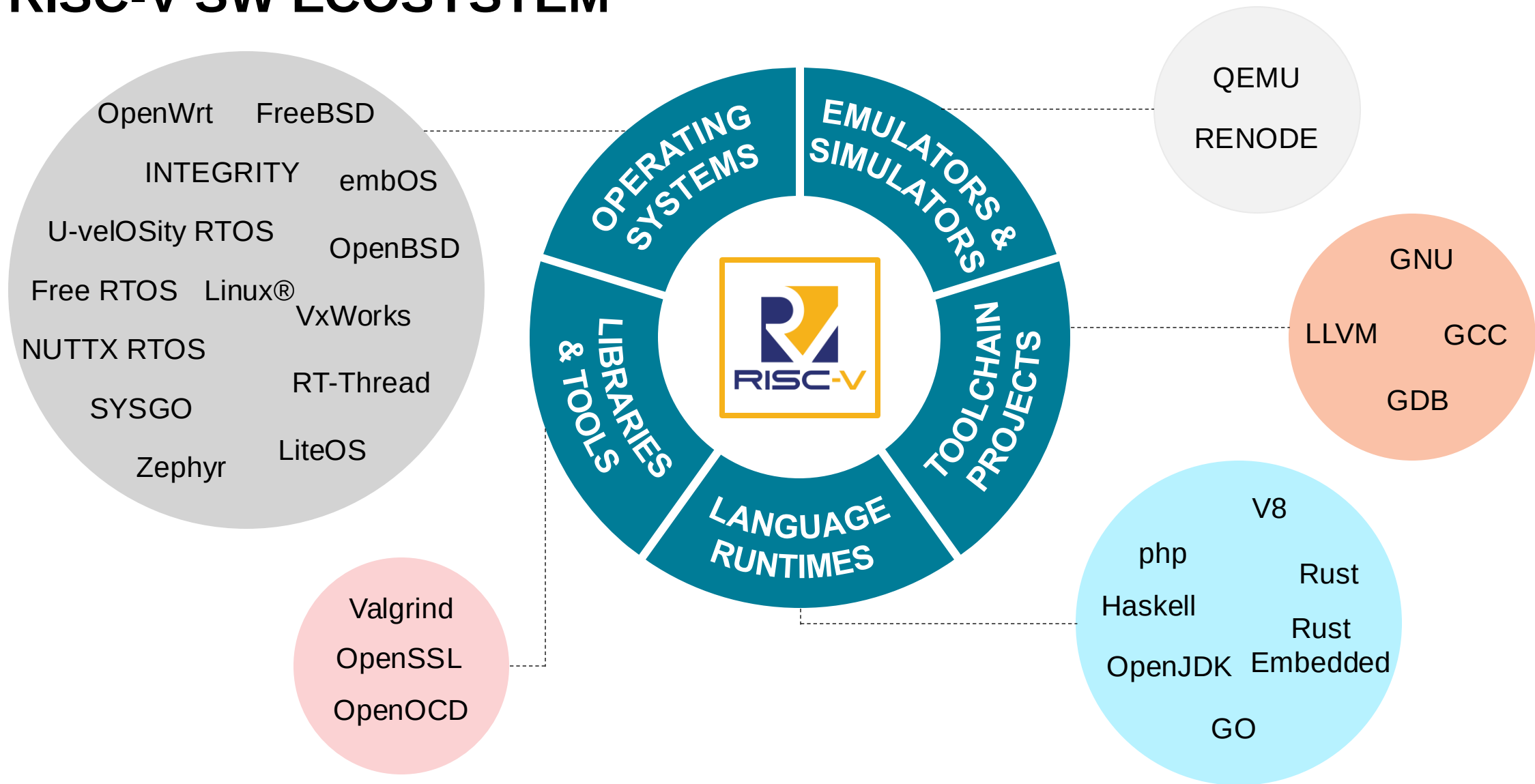
Flexible and Extensible Instruction Set

- Customizable for specific workloads
- Adaptable to evolving applications and workloads like vector processing, machine learning, or security extensions

WHY RISC-V?



RISC-V SW ECOSYSTEM



RISC-V INSTRUCTION SET EXTENDABILITY

Name	Description
RV32I/RV64I	Base integer instruction set examples
A	Extension for Atomic instructions
C	Extension for Compressed instructions
F & D	Extensions for single and double precision floating-point instructions
H	Extension for hypervisor instructions
M	Extension for integer multiplication and division instructions
S/U	Extensions for Supervisor/User mode instructions
V	Extension for vector instructions
Zkne/Zknd	Extensions for AES encryption and decryption instructions
Zknh	Extension for SHA2 hashing instructions

The RISC-V ISA is comprised of base instructions and extensions

The RISC-V ISA extensions define optional components that a processor may implement

There are 33 ratified extensions and many others under development

Some popular extensions are listed in this table

AMD ADAPTIVE SOC AND FPGA SOFT PROCESSOR LANDSCAPE



- 20-year heritage FPGA & AMD Zynq™ SoC deployment
- AXI based, leveraging Arm® compatible portfolio of IP
- Custom RISC instruction set with custom gcc
- Reliable mature technology
- Highly integrated tools
- Simple design methodology



Builds on MicroBlaze experience

AXI support

RISC-V ISA implementation. Compatible with open-source gcc

First release in 2024.1

Highly integrated in tools

Follows MicroBlaze design methodology

Compatible with a growing ecosystem of tools, firmware and OS

At this stage, there is no plan to deprecate AMD MicroBlaze processor support

EMBEDDED PROCESSING ON AMD FPGAS AND ADAPTIVE SOCS



Fast soft-processor performance with the highest flexibility

**Programmable
Systems Integration**



FPGA provides flexibility to adapt to evolving ISA extensions

**BOM
Cost Reduction**



Wide range of optimized IPs available to deploy an AMD MicroBlaze™ V core subsystem in PL

Power Savings



A reduced BOM reduces power consumption

**Accelerated Design
Productivity**



Configurations are highly scalable across a wide range of applications

AMD MICROBLAZE™ V PROCESSOR TURNS DEVICES INTO EMBEDDED SYSTEMS

Lock-step Capable

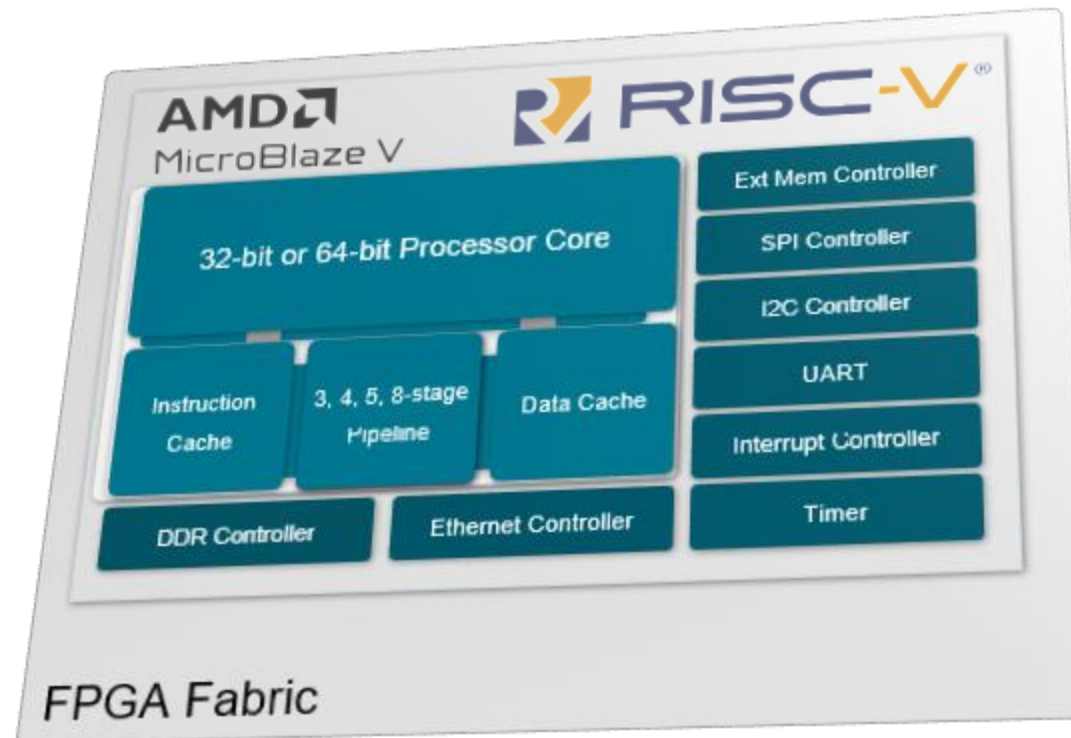
- Tamper resistance
- Fault protection

Triple Modular Redundant Capable

- Single Event Upset mitigation
- Voter circuit

Microcontroller Subsystem

- Area optimized
- Integrated with peripherals



Multiple Configurations

- Configurable for area & performance
- Support for multiple ISA extensions

AXI Interconnect

- Plug-and-play IP
- Standards compliant

FPGA Fabric

- Custom engines
- Extendable peripherals

AMD MICROBLAZE™ V PROCESSOR CONFIGURATION OVERVIEW

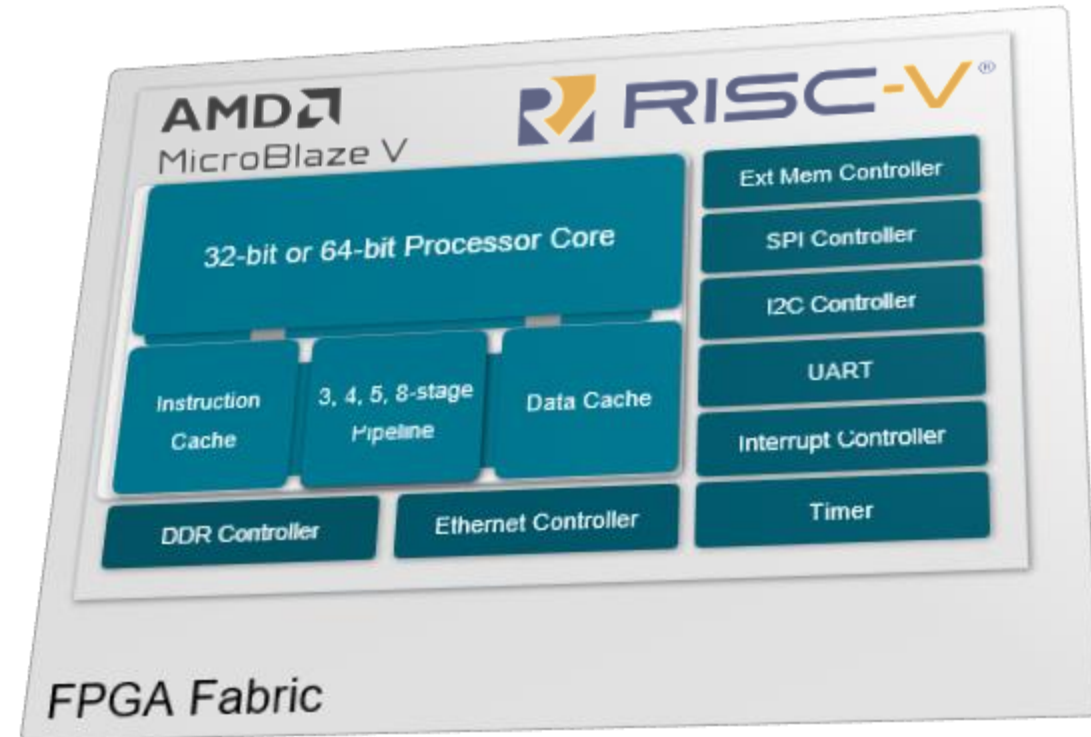
Available as:

- Microcontroller
- Memory Protection Unit for physical memory protection (EA in 2024.1)
- *Memory Management Unit (MMU) for memory virtualization (EA in 2024.2)*

Configurable

- RISC-V ISA support
- Pipeline length (3, 4, 5, or 8 stage)
- Instruction and data caches

Like many of our IP, the RTL source code is encrypted



Licensed as part of AMD Vivado™ Design Suite on any AMD device at no extra cost

AMD MICROBLAZE™ V PROCESSOR: HARDWARE OVERVIEW

Local bus, Full AXI, and Streaming AXI interfaces

Large library of Arm® compatible AXI bus soft peripherals

Debug support using AMD MicroBlaze™ Debug Module V
RISC-V trace support coming soon

Configurable Example Designs

- AMD MicroBlaze V Presets
- AMD Versal™ MicroBlaze V Presets

AMD MicroBlaze Micro Controller Subsystem V (MCS V)

- Highly integrated & area optimized processor for controller applications
- Common peripherals like UART, Timers, GPIO, and Interrupts



AMD MICROBLAZE™ V RISC-V ISA SUPPORT

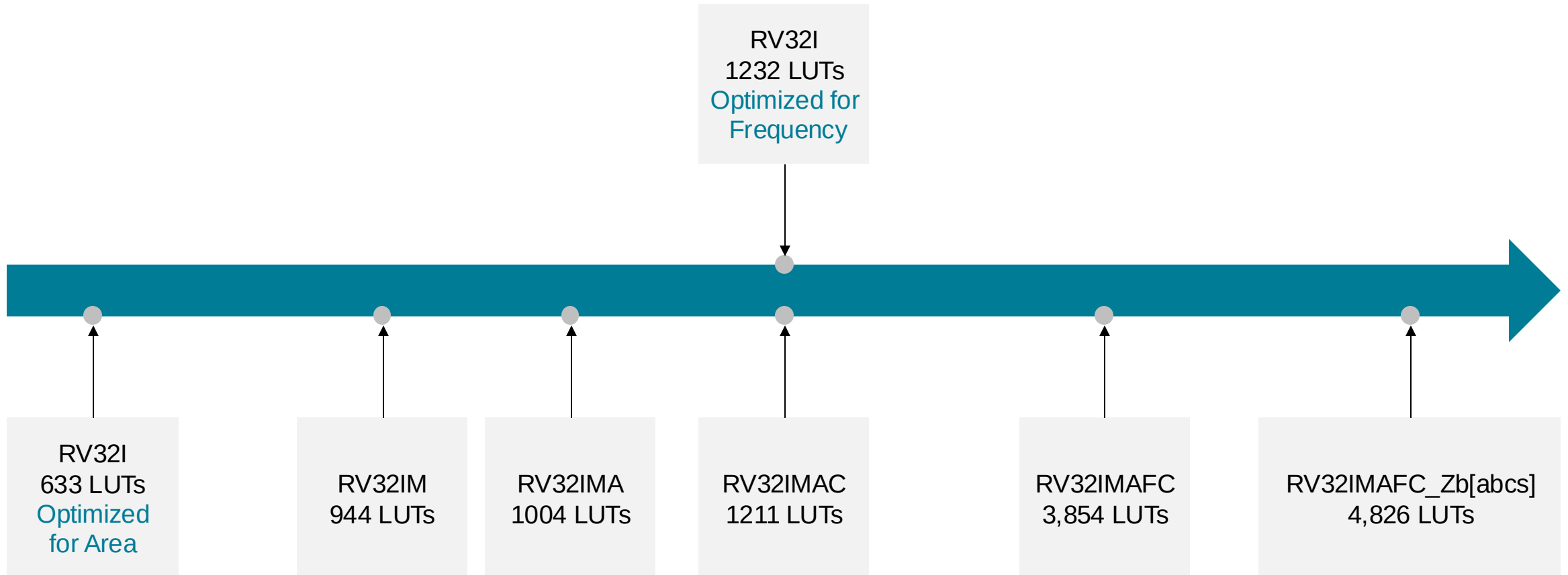
Name	Description
RV32I	32-bit base integer instruction set
RV64I	64-bit base integer instruction set (EA in 2024.1)
M	Extension for integer multiplication and division
A	Extension for atomic instructions
F	Extension for single-precision floating-point
D	<i>Extension for double-precision floating-point (EA in 2024.2)</i>
C	Extension for compressed instructions
Zb[abcs]	Extensions for bit manipulation (Zba, Zbb, Zbc, Zbs)
Zicsr	Extension for control & status register (CSR)
Zifencei	Extension for instruction-fetch fence
Zicbom	Extension for cache block management

Configurable ISA support lets the designer choose whether operations are implemented in hardware (for speed) or emulated in software (less logic)

A similar concept to floating-point acceleration but at a much more granular level

We are interested in your feedback

BUILD THE RIGHT PROCESSOR FOR YOUR APPLICATION



Synthesized for AMD Spartan™ 7 FPGA – not drawn to scale

AMD TOOLS FLOW FOR MICROBLAZE™ V PROCESSOR

AMD 
Vivado

Hardware Design

- Configure the optimal processor for your application using development flow of choice
 - RTL Flow
 - IP Integrator Flow
- Export processor and peripheral configuration to AMD Vitis™ tools

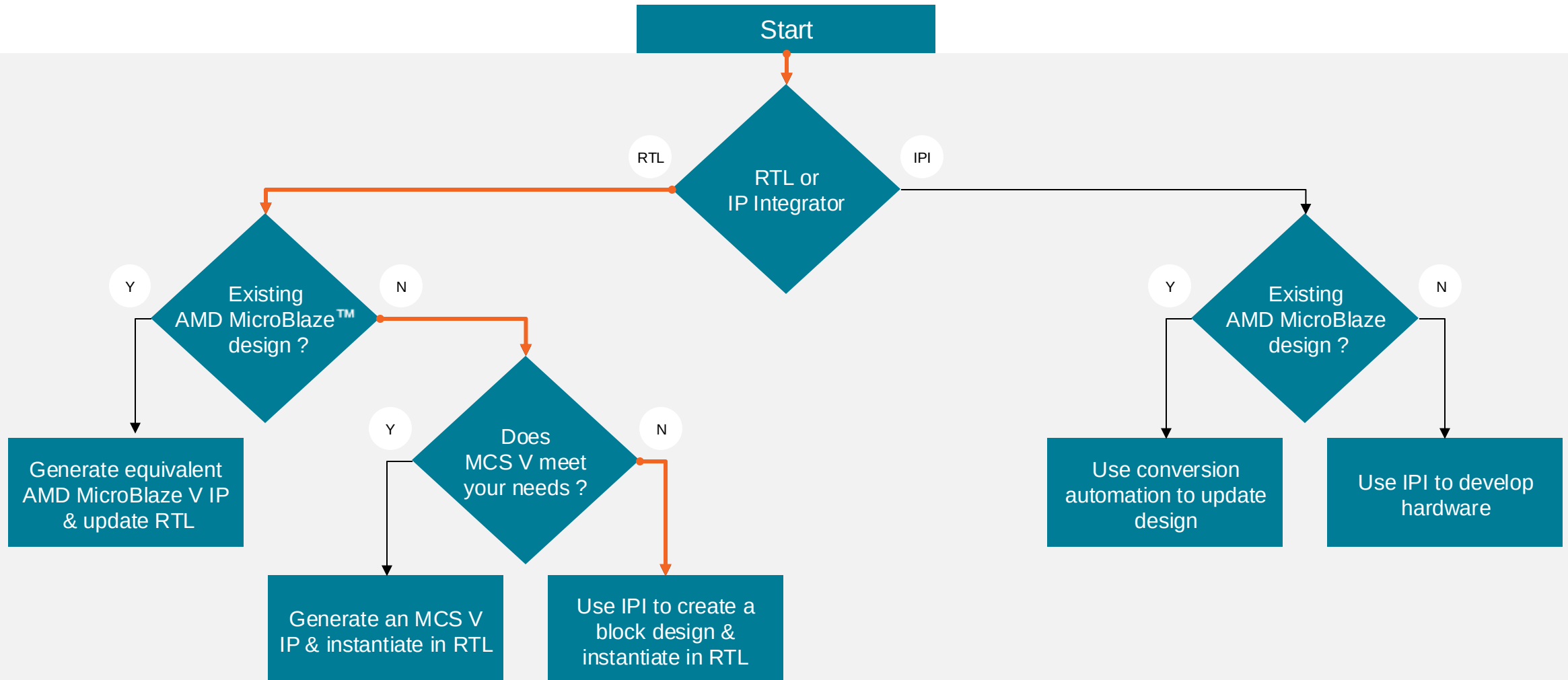


AMD 
Vitis

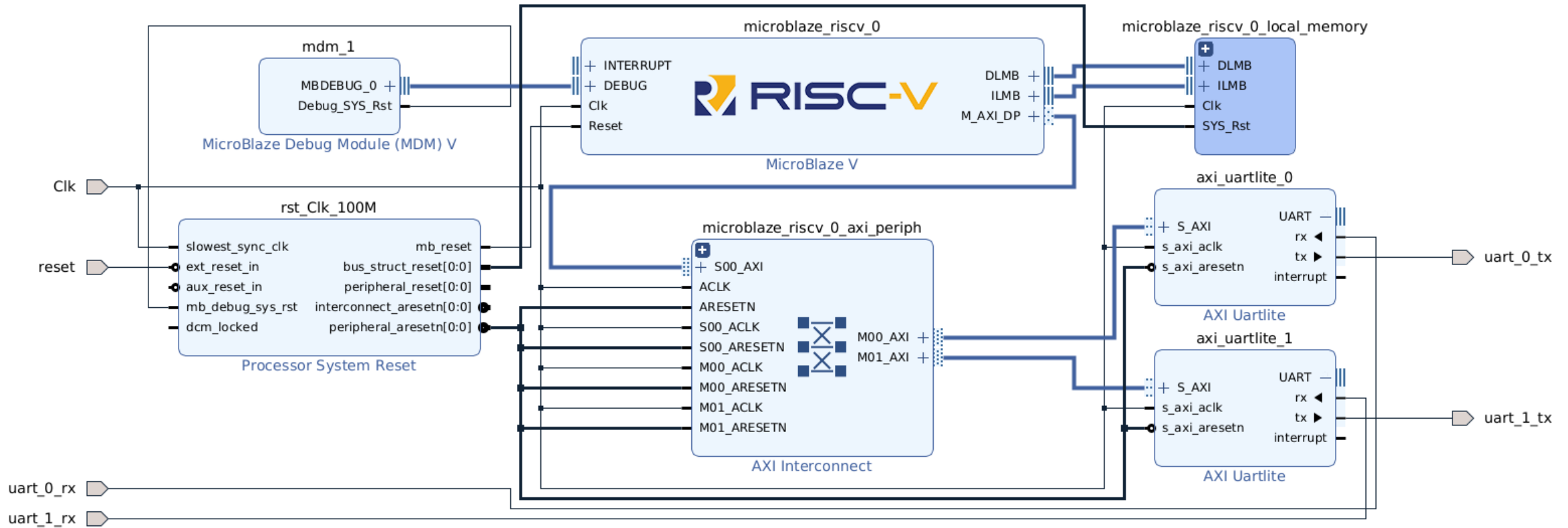
Software Development

- Develop, build, run & debug your software
- When design is ready, update hardware design with software image

SUGGESTED DESIGN FLOW FOR AMD MICROBLAZE™ V PROCESSOR



ADVANTAGES OF USING IP INTEGRATOR – EXAMPLE



Create designs by
interconnecting IP cores in a GUI

Automations make
interconnection easy

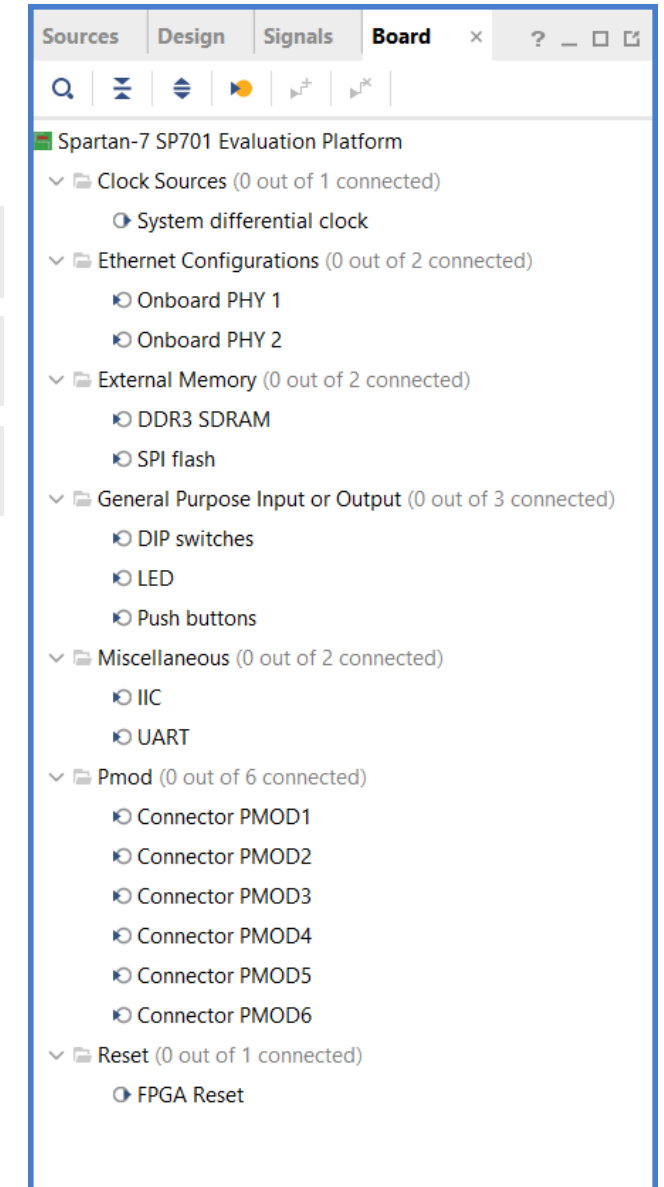
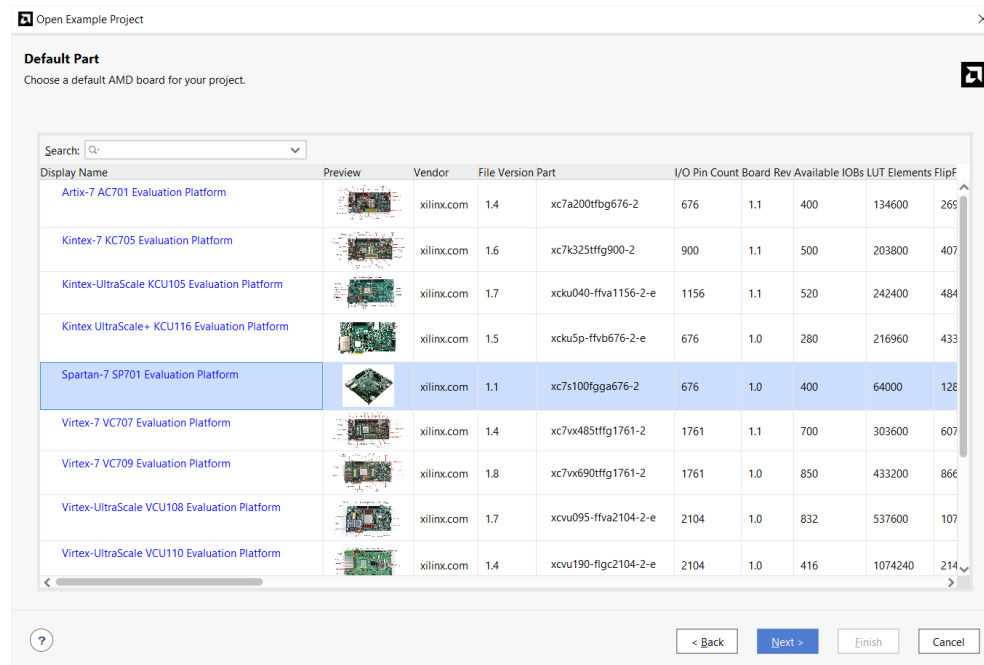
Validity checks ensure design
integrity

AMD VIVADO™ DESIGN SUITE – BOARD SUPPORT

Example designs for multiple AMD and partner boards

Drag-and-drop board interfaces within IP Integrator

IP Integrator automatically adds IP to match interface and connects it to the AMD MicroBlaze V Processor



AMD MICROBLAZE™ V PROCESSOR DEBUG MODULE (MDM V)

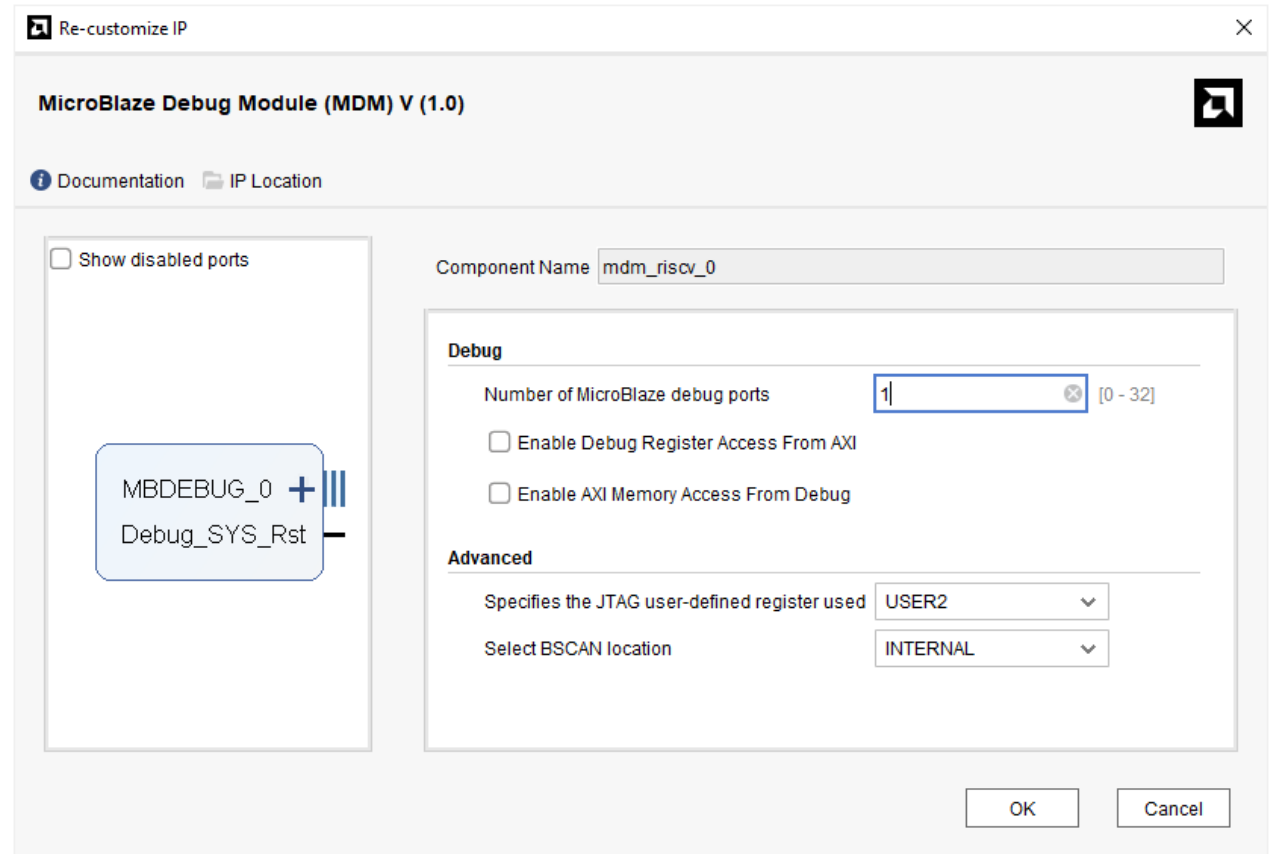
Debugs up to 32 AMD MicroBlaze™ V cores

Enables JTAG-based debugging of one or more MicroBlaze V processors

Direct JTAG-based access to memory with a configurable AXI4 master interface

Based on boundary scan (BSCAN) logic in AMD FPGAs & adaptive SoCs

RISC-V Debug Support Version 1.0.0-STABLE



AMD MICROBLAZE™ V PROCESSOR – FAMILIAR SOFTWARE FLOW



AMD MicroBlaze™ V uses the same Vitis™ software platform as other AMD processors

AMD Vitis tools contain a complete development package for developing applications

AMD Vitis tools use the GCC 12.2 toolchain in 2024.1

We are talking to partners about third-party AMD MicroBlaze V processor support

Zephyr RTOS support coming soon

Linux® support coming soon

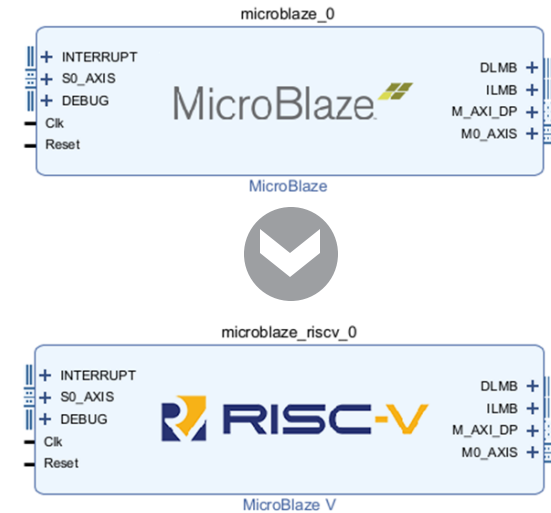
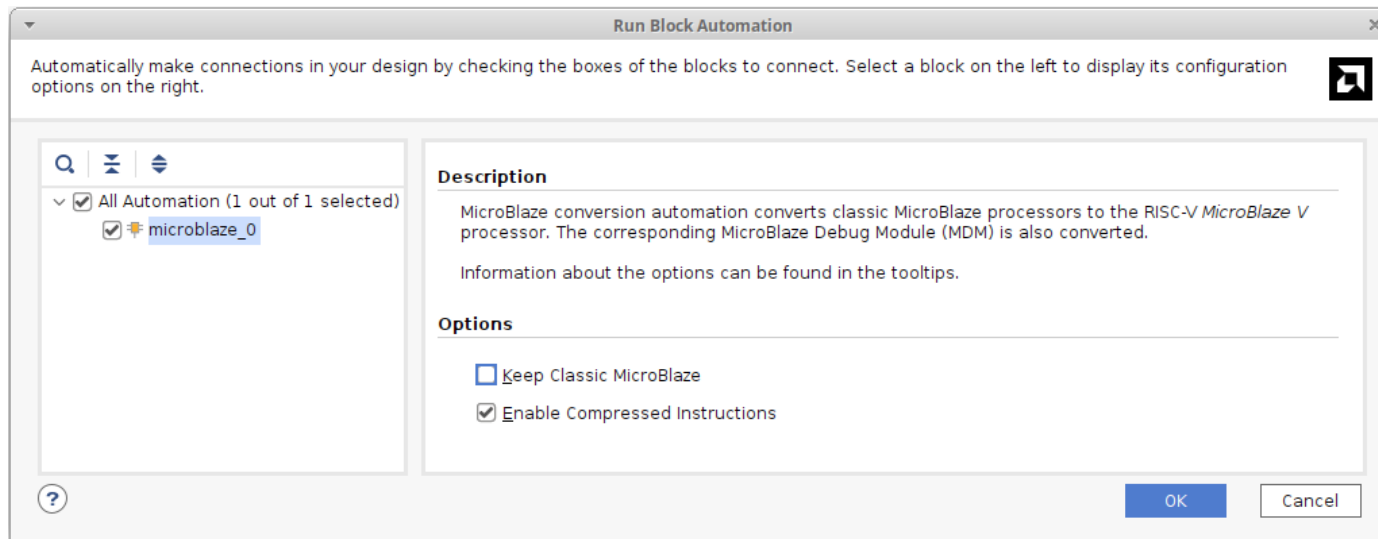


AMD MICROBLAZE™ V FOR MICROBLAZE USERS

Same development flow as the AMD MicroBlaze™ processor

Same interfaces and peripheral support as the AMD MicroBlaze processor

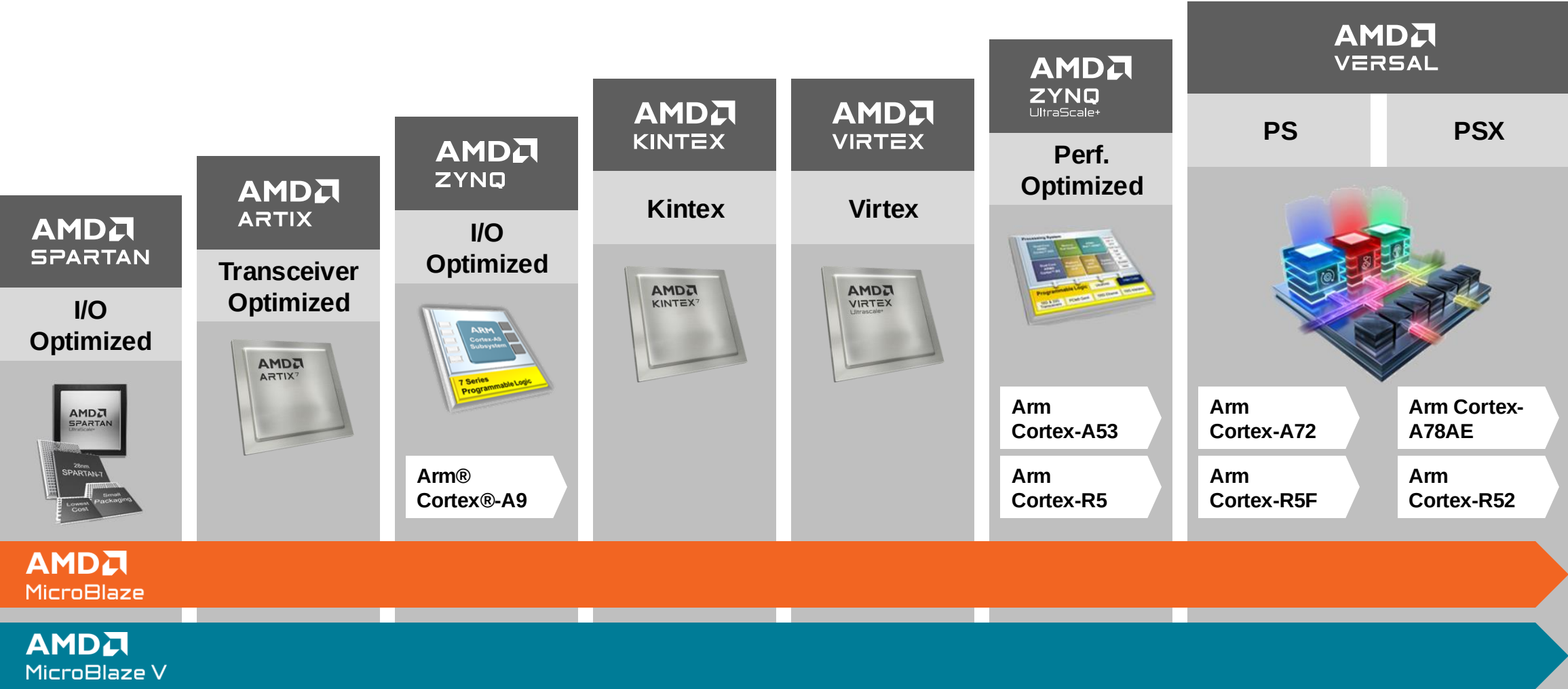
Automation to convert IP Integrator MicroBlaze designs to AMD MicroBlaze V (AMD Vivado™ 2024.1)



At this stage, there is no plan to deprecate MicroBlaze processor support

For new designs, highly recommended to use MicroBlaze V processor

AMD EMBEDDED PROCESSOR PORTFOLIO

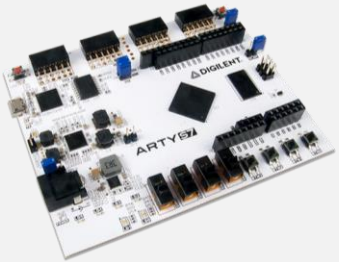


EXPLORE AMD MICROBLAZE™ V WITH COST-OPTIMIZED DEVELOPMENT KITS

AMD
SPARTAN7

Arty S7

Hobbyist Starter Kit



7S25

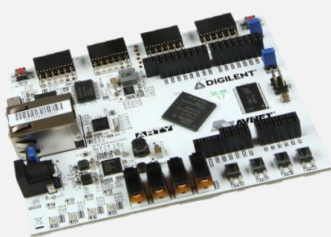
[Available Now](#)



AMD
ARTIX7

Arty A7

Hobbyist Starter Kit



7A100T

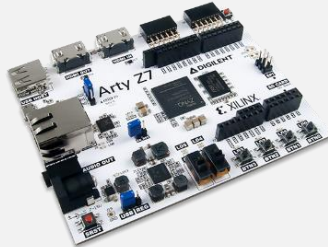
[Available Now](#)



AMD
ZYNQ
7000

Arty Z7

Hobbyist Starter Kit



7Z7010

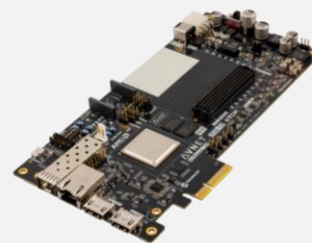
[Available Now](#)



AMD
ARTIX
UltraScale+

AUBoard 15P

Development Kit



AU15P

[Coming Soon](#)



AMD
ZYNQ
MPSoC

ZUBoard 1CG

Development Kit



ZU1CG

[Available Now](#)



Ultra96-V2

Embedded Linux®
Starter Kit



ZU3EG

[Available Now](#)



KEY TAKEAWAYS



➤ Cost-optimized RISC-V processor tailored for your specific design requirements



➤ Processing scalability from AMD MicroBlaze™ V soft cores to single-core or multi-core heterogeneous Arm® processing



➤ IP-centric & system-centric design environments that are easy to use and enable scalable HW / SW development

CALLS TO ACTION



Use AMD MicroBlaze V Configurable Example Designs on an AMD or Technology Partner evaluation board



Refer to UG1629 and PG428 for more details (available with 2024.1)

DISCLAIMER AND ATTRIBUTIONS

DISCLAIMER

Timelines, roadmaps, and/or product release dates shown in these slides are plans only and subject to change.

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